OPENADMET CYP INHIBITION CHALLENGE

Training on the metric, and recovering the inactives

Our project builds a multitask message passing network that predicts direct inhibition of four CYP enzymes: CYP1A2, CYP2C9, CYP2D6 and CYP3A4. The loss function is the same as the competition metric, and the training set includes compounds that the dose response data leaves out.

Track: direct inhibition, regression Metric: macro averaged STRAE Framework: chemprop 2.2.3 Proprietary data: none

SUMMARY

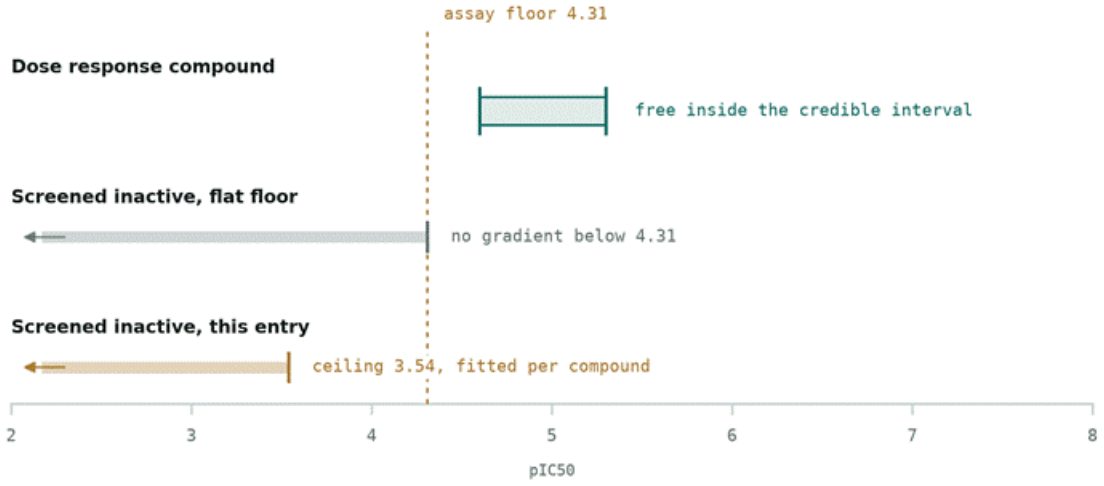
What our entry does differently

Our entry rests on three main ideas. We tested each one in the ablation study instead of just assuming it helps.

- **The loss function is the metric.** STRAE gives no penalty when a prediction lands inside a compound's credible interval. We found that this error splits exactly into two one sided censored losses, and chemprop already has both built in, so the loss we train on is the same function the leaderboard uses.
- **The dose response set is a hit list, so we recover the inactives.** A compound only has a pIC50 if it already looked active, so the dose response data leaves out the compounds that did nothing. The 17,504 single concentration measurements tell us which compounds were screened and did not inhibit. We feed these into the same bounded loss as inequalities.
- **Each inactive gets its own upper bound.** If we censor every inactive at the assay floor, we throw away the graded signal in the screen, and the model has no reason to predict anything lower. Instead we regress log2 fold change onto pIC50, which gives each inactive a bound of its own.

SCORING GEOMETRY

Three kinds of label on one scale



The metric does not penalise anything that falls inside the interval, so the interval is what we train on. With a flat floor the model stops at 4.31, because the bounded loss charges nothing below the censoring value. The fitted ceiling moves that boundary to where the screen says the compound really sits.

DATASET

Sparse labels, a dense test set

We used five files from the challenge release. Only the first two make up the main pair of datasets; the rest give extra signal.

TRAINING LABELS BY ISOFORM

Isoform	Dose response	Recovered inactive	Total usable	Median interval
CYP1A2	1,412	2,499	3,911	0.33
CYP2C9	1,285	1,276	2,561	0.53
CYP2D6	1,493	2,725	4,218	0.27
CYP3A4	2,335	1,213	3,548	0.38

The training matrix is only 33.3 percent filled in: 6,525 measured cells out of 19,620 possible, and 73.3 percent of compounds have just one isoform label. The blinded test set is completely filled in, with 750 compounds across all four isoforms. This mismatch explains most of the choices we made below.

The test set is also an analog expansion and not a random sample, so it is closer to the training set than the training set is to itself (median maximum Tanimoto similarity 0.587 versus 0.449). This means a scaffold split would be harder than the real leaderboard problem and would lead us to pick the wrong model.

FEATURES

A learned graph plus 31 bulk properties

The network learns its own representation from the molecular graph. We then attach 31 RDKit descriptors to that learned fingerprint. CYP inhibition depends strongly on bulk properties, and without the descriptors the message passing layers would have to rediscover this from limited data.

Standard set, 20 descriptors

Molecular weight, MolLogP, TPSA, hydrogen bond donors and acceptors, rotatable bonds, aromatic and aliphatic ring counts, ring count, fraction of carbon atoms that are sp3, heavy atom count, heteroatom count, molar refractivity, Bertz complexity, Labute surface area, Balaban J, Hall Kier alpha, saturated rings, NHOH count, NO count.

Shape and flatness, 6 descriptors

Aromatic proportion, Kappa 1 through 3, Chi1v, amide bond count and QED. Aromatic proportion can tell a flat biaryl apart from a puckered saturated system with the same ring count, which plain ring counts cannot do. It correlates 0.30 to 0.33 with potency at CYP1A2, CYP2C9 and CYP3A4.

Ionisation, 5 descriptors

Basic nitrogen count, basic nitrogen adjacent to an aromatic ring, strictly protonatable nitrogen, and logD at pH 7.4. CYP2D6 recognises a nitrogen that is protonated at physiological pH and forms an ion pair with Glu216 and Asp301, and the standard descriptor set has no basicity term at all. The strict count, which includes aliphatic amines but excludes anilines and amides, correlates +0.302 with CYP2D6 and between −0.10 and −0.19 with the other three. logD behaves the same way: it correlates −0.180 with CYP2D6, while raw logP gives +0.034. CYP2D6 prefers a charged molecule and the other three prefer a greasy one, so the two measures have opposite signs.

> We also tested the topological distance from the basic nitrogen to the nearest aromatic ring, and we rejected it. It seems to correlate +0.266 with CYP2D6, but it is zero for 82 percent of compounds, so it really just re-detects whether a basic nitrogen is there. Among compounds that have one, the correlation is −0.171, and the binned means fall steadily instead of peaking at any particular distance.

PREPROCESSING

Scaling, and the inactive call

Scaling

Descriptors and targets are standardised using the training split only. In chemprop, the scale transform does nothing in training mode and is applied at inference. So we normalise the training dataset in place, while every evaluation path gets raw values and is scaled once by the model itself. Normalising the validation or prediction dataset as well would scale the data twice, which shrinks the descriptor spread by a factor of fifty and pushes every value far outside the range seen in training.

Deciding what counts as inactive

Statistical significance turned out to be the wrong test. At 49.5 micromolar with tight replicate errors, 92 percent of never promoted CYP3A4 compounds pass a false discovery rate below 0.05, even though the effects are far too small to matter. So we calibrated an effect size threshold for each isoform instead. Using

compounds that have both a screen measurement and a dose response value, we took the least negative log2 fold change at which at most 10 percent of compounds turn out to be active. This gave thresholds of −0.75, −0.44 and −1.82 for CYP1A2, CYP2C9 and CYP3A4. CYP2D6 has no usable threshold, because its mass spectrometry readout has a compressed range and its calibration subset is 85 percent active. For CYP2D6 we used the significance test instead, and the notebook says so openly rather than inventing a number.

Where an inactive actually sits

log2 fold change correlates between −0.70 and −0.89 with dose response pIC50. We therefore regress one on the other, which turns each screen reading into an estimated pIC50 and gives every inactive its own censoring ceiling. The medians drop from a flat 4.31 to 3.73, 4.00, 3.54 and 3.96. The bound stays one sided, which is a weaker claim than the regression point estimate and so fails safely. It also carries a margin of one residual standard deviation and is clamped at each isoform's weakest measured value.

MODEL

Architecture

- **Encoder.** A bond level directed message passing network over the molecular graph, with mean aggregation and batch normalisation. It has roughly 227 thousand parameters.
- **Head.** One feed forward regression head with four outputs, one for each isoform. It takes the learned fingerprint plus the 31 descriptors and has roughly 100 thousand parameters.
- **Multitask by design.** 27 percent of training compounds have more than one isoform label, and the four enzymes share a lot of recognition chemistry. CYP2D6 has only 1,493 dose response labels, so it benefits especially from the shared encoder.
- **Ensemble.** We average five seeds. A mean variance estimation head gives an uncertainty for each prediction, which we use for diagnostics and not for the submission.

The loss

The STRAE error for each compound is the sum of two hinges: one above the upper bound of the credible interval and one below the lower bound. Taken separately, the first is the loss for a right censored observation and the second is the loss for a left censored one. chemprop implements both as bounded mean absolute error with per target inequality masks, which are meant for assay readouts recorded as greater than or less than some concentration. So each compound goes into training twice, once bounded below at the lower bound and once bounded above at the upper bound, and the two losses add up to the metric term for term. We needed no custom loss and no custom trainer, and the validation loss becomes a direct proxy for the competition score instead of just being correlated with it.

TRAINING

Process and model selection

- **Optimisation.** We used the Lightning trainer with batch size 64 and up to 40 epochs, with early stopping on validation loss (patience 8). Models are cached on disk under a key made from the configuration, descriptor count and a version token, so a change in training behaviour cannot quietly reload an old checkpoint.

- **Validation split.** We built it to match the similarity profile of the test set. The test to train maximum Tanimoto histogram acts as a quota, and each compound we select has its three nearest neighbours protected from also moving into validation. Without that protection, the matched similarity falls back to roughly random.
- **Validation labels.** The fold only scores cells that were promoted to dose response, and promotion is the hit filter. That makes the fold 86 percent active for CYP2D6, while a randomly screened compound is active only about 29 percent of the time. We fill the missing cells with the screen's inactive calls, which brings CYP2D6 down to 27 percent active. Censored records that belong to validation compounds are dropped from training.
- **Selection.** We tried six configurations, each averaged over three seeds, and chose between them on the corrected scorer. The spread across single seeds is about 0.015 and the configurations are only roughly 0.02 apart, so picking on one seed would be close to a coin flip at the step that decides the submission.
- **Final fit.** We refit five seeds on all 4,905 training compounds. Each one holds out a different random five percent for early stopping, which also adds diversity to the ensemble.

RESULT

Ablation and leaderboard

ABLATION STUDY ON THE HELD OUT FOLD, MACRO STRAE. EACH ROW ADDS ONE INGREDIENT TO THE ROW ABOVE, EXCEPT F, WHICH REMOVES ONE.

Configuration	CYP1A2	CYP2C9	CYP2D6	CYP3A4	Macro
A point target, mean squared error	1.5668	0.6686	1.6536	0.6410	1.1325
B soft threshold interval loss	1.2774	0.6581	1.8095	0.5960	1.0853
C plus recovered inactives	0.8975	0.5643	0.6709	0.5505	0.6708
D plus RDKit descriptors (submitted)	0.7367	0.5208	0.6905	0.5462	0.6236
E plus pretrained encoder	0.9480	0.5253	0.6966	0.4773	0.6618
F inactives removed	1.7060	0.7645	2.0552	0.6545	1.2951

The biggest effect comes from the recovered inactives. Going from configuration B to C moves the macro score from 1.0853 to 0.6708, and CYP2D6 alone goes from 1.8095 to 0.6709. Row F tests the same ingredient by removing it, and it is the worst of all six. Averaging five seeds of the selected configuration reaches 0.5809 on the held out fold.

LEADERBOARD RESULTS FOR EACH ISOFORM, FROM THE MOST RECENT SCORED SUBMISSION

Isoform	STRAE	MAE	R squared	Spearman rho
CYP1A2	0.5404	0.795	0.4922	0.7767
CYP2C9	0.5079	0.5071	0.5807	0.7867
CYP2D6	1.0424	1.5099	−0.4209	0.5078
CYP3A4	0.5540	0.592	0.5969	0.7941

Overall, macro STRAE is 0.6612, macro R squared 0.3122, macro Spearman rho 0.7163 and macro Kendall tau 0.5307.

Three of the isoforms score within 0.05 of each other. CYP2D6 on its own accounts for 39 percent of the total error, and if it matched the others the macro score would be close to 0.538. Our per compound censoring ceiling, described above, is meant to fix this, but it is not yet reflected in a scored submission. On the held out fold it moves the CYP2D6 predictions from a mean of 4.38 (standard deviation 0.40) to a mean of 3.80 (standard deviation 0.82), compared with an independently implied true mean near 2.9.

Time dependent inhibition

We also built a separate multitask classifier with the same encoder design. It takes substructure alerts for mechanism based inhibition as extra inputs. We tune the thresholds on the validation fold instead of leaving them at 0.5, which is worth 0.02 to 0.05 Matthews correlation when positive rates are near 21 percent. Validation Matthews correlation reaches 0.4299 for CYP3A4 (area under the curve 0.8128) and 0.3669 for CYP2D6 (area under the curve 0.6678).

REPRODUCIBILITY

What you need to rerun it

Everything runs from one notebook in one environment, and no proprietary data is needed. The dependencies are chemprop 2.2.3, RDKit, PyTorch Lightning and scikit learn. The public CYP inhibition benchmark from the Therapeutics Data Commons is used only for optional encoder pretraining in configuration E, which we did not select. We copied the scoring functions from the organisers’ reference implementation, so the quantity we optimise is exactly what the leaderboard computes.